Career Pathways and the Future of Low-Code Development

Project: Insurance Claim Management System

Experiment No	15
Name	Adishwar Seelan
Roll No	241401004
Department	CSBS

1. Aim

To study the career pathways open to low-code developers and the likely future of low-code development across the industry — not only on Zoho Creator, but on low-code and no-code platforms in general — and to prepare a report and presentation outline that connects these trends to the skills built in the earlier experiments.

The study helps in planning which skills to build next and which roles to aim for as low-code, pro-code and AI-assisted development grow closer together.

2. Procedure

Step 1: Trace the evolution — The development of application building from hand coding to low-code platforms and AI-assisted building was outlined as a timeline (Section 3.1).

Step 2: Survey the platform landscape — Well-known low-code platforms were compared by focus and typical use, to show that the field is wider than any single product (Section 3.2).

Step 3: Identify future trends — Current industry trends such as AI-assisted building, AI agents, governed citizen development, hyperautomation and legacy modernisation were collected from general industry knowledge and summarised with their effect on careers (Section 3.3), and the likely future workflow was drawn (Section 3.9).

Step 4: Map career pathways — Roles were arranged from entry to senior level, with possible switches to other careers, as a pathway diagram and a role table (Sections 3.4 and 3.5).

Step 5: Build the skill stack and roadmap — The skills needed were arranged in layers and turned into a step-by-step learning roadmap using the experiments already completed as the starting point (Sections 3.6 and 3.7).

Step 6: Examine risks — Challenges such as shadow IT, security, lock-in and the quality of AI-generated output were listed with ways to handle them (Section 3.8).

Step 7: Prepare the presentation — A slide-by-slide outline was drawn up so the findings can be presented clearly (Section 3.10).

Note: *this is a study based on general industry knowledge. No forecast figures are quoted; any statistics found in articles come from secondary sources and should be verified before being used as facts. The timeline and roles are a general guide, not a prediction for any one company.*

3. Output

3.1 Evolution of application development

The timeline shows the move from hand coding to rapid-development tools, then visual low-code platforms, citizen development with hyperautomation, and now AI-assisted and agent-based building. Each step reduced the effort needed to produce an application while adding the need for oversight.

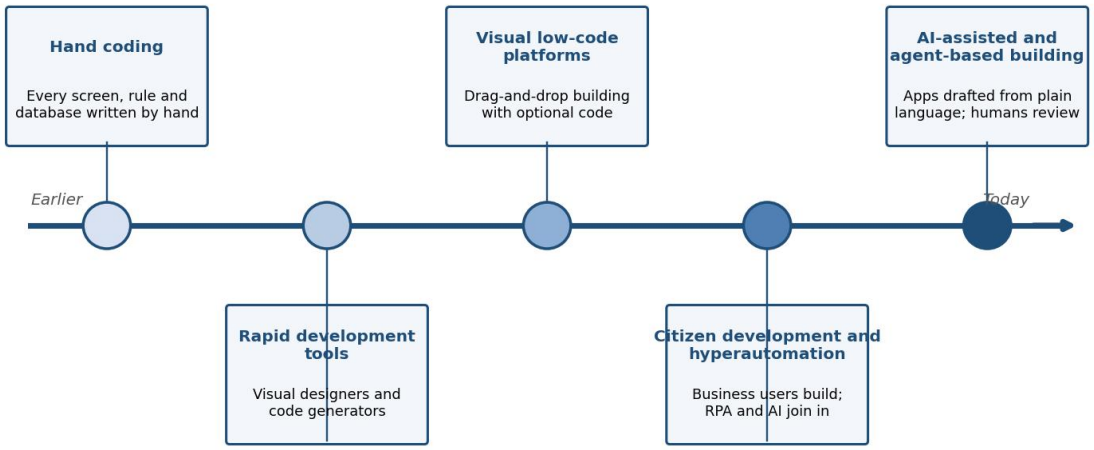


Figure 1: Evolution of Application Development (Conceptual Timeline)

3.2 The platform landscape beyond Zoho Creator

Low-code is a broad field. The platforms below are examples of different focus areas; the skills transfer between them.

Platform	General focus	Typical use
Zoho Creator	Business apps with strong links to the Zoho ecosystem.	Small and mid-size business process apps, such as the claim management system built in these experiments.
Microsoft Power Apps	Apps tied to the Microsoft 365 and Azure ecosystem.	Internal tools for organisations already using Microsoft services.
OutSystems	Enterprise-scale development with strong code extensibility.	Large, complex enterprise applications.
Mendix	Enterprise application platform with visual modelling.	Enterprise and industrial applications.
Appian	Process automation and case management.	Workflow-heavy regulated processes, such as claims handling.
Salesforce Platform	Custom apps built around CRM data.	Customer-centred business applications.

3.3 Trends shaping the future

Trend	What it means	Effect on careers
AI-assisted building	Apps and logic can be drafted from plain-language descriptions.	Less time on routine building, more on reviewing, design and testing.
AI agents in applications	Agents act inside workflows, not only suggest.	New roles in designing, limiting and monitoring automated agents.
Governed citizen development	Business users build apps under IT rules.	Demand for platform owners and centres of excellence to set standards.
Hyperautomation	Low-code, RPA and AI combined to automate whole processes.	Automation engineers and process designers are valued.
Legacy modernisation	Old systems are wrapped or replaced using low-code.	Integration and migration skills become important.
Fusion teams	Business and technical staff build together.	Communication and requirement skills matter as much as tools.

3.4 Career pathways

The diagram shows common routes from entry roles such as citizen developer or junior low-code developer, through mid-level roles in development, integration and automation, to senior roles such as architect, platform lead or AI-automation architect. Dashed arrows show switches to pro-code engineering, product ownership or entrepreneurship.

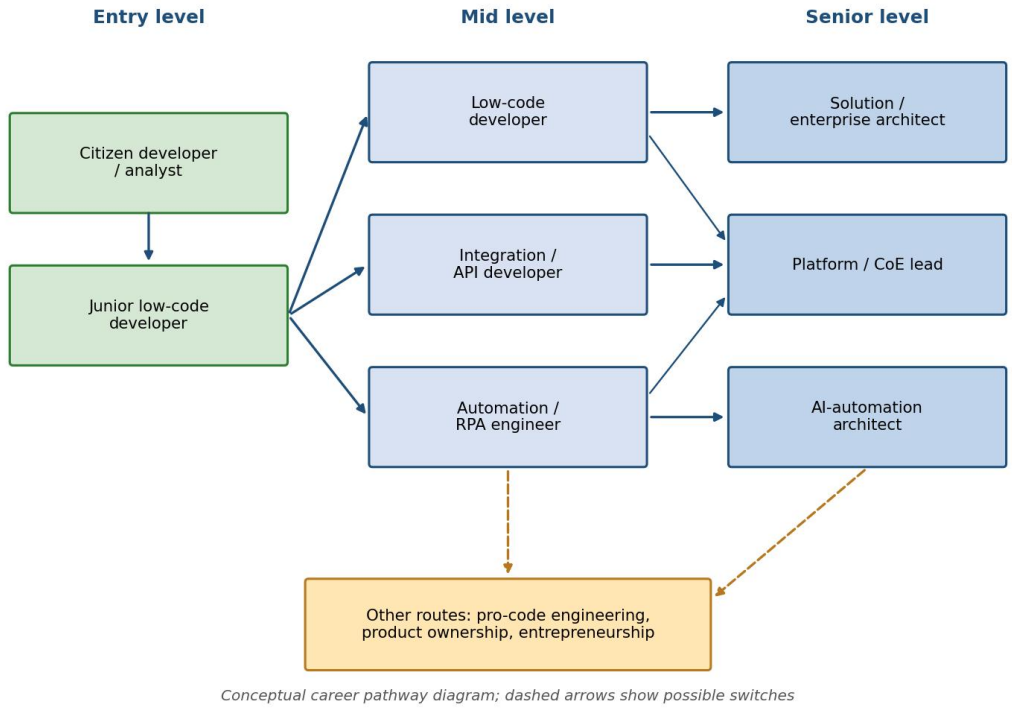


Figure 2: Career Pathways in Low-Code Development (Conceptual Diagram)

3.5 Roles at a glance

Role	What the person does	Key skills
Citizen developer / analyst	Builds simple apps for own team; gathers requirements.	Process thinking, basic platform use.
Low-code developer	Builds forms, workflows, reports and scripts.	Platform skills, data modelling, scripting.
Integration / API developer	Connects apps to other systems through APIs.	REST, JSON, authentication, error handling.
Automation / RPA engineer	Automates end-to-end processes, including AI steps.	Workflow design, RPA tools, AI services.
Solution / enterprise architect	Designs overall solution, security and governance.	Architecture, security, platform strategy.
Platform / CoE lead	Sets standards and supports citizen developers.	Governance, training, communication.

3.6 Skill stack for a low-code career

Skills are arranged in layers from foundations such as logic and data to platform, integration, pro-code extension, AI and automation, and governance. Each layer adds value on top of the one below, and the top layers are the ones least likely to be replaced by automation.

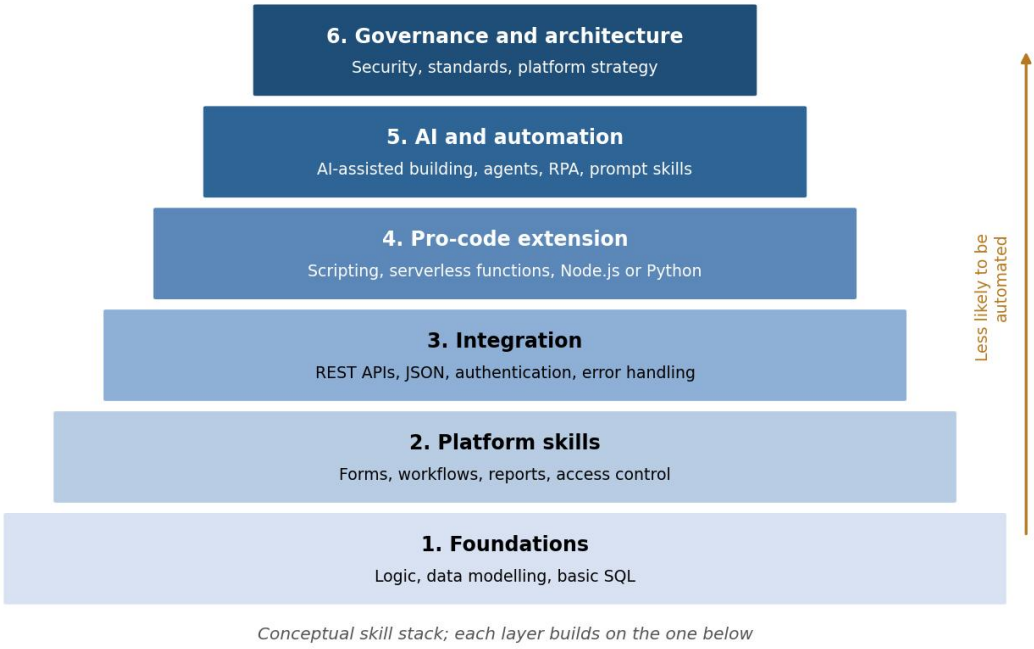


Figure 3: Skill Stack for a Low-Code Career (Conceptual Diagram)

3.7 Learning roadmap

The roadmap moves from basics to projects, then to connections and scripting, AI skills and finally specialisation. The experiments completed so far already cover steps two and three: the Insurance Claim Management System, Deluge workflows, role-based access, the Zoho Calendar connection and pro-code extension.

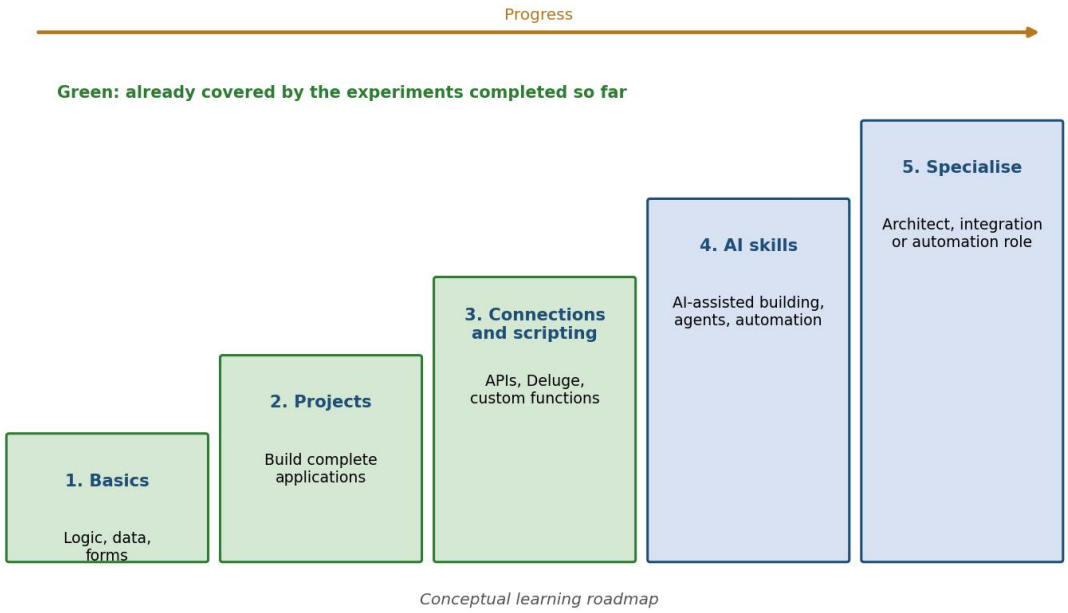


Figure 4: Learning Roadmap (Conceptual Diagram)

3.8 Risks and challenges

Challenge	Why it matters	How to handle it
Shadow IT	Apps built outside IT may be unknown, unsecured or duplicated.	Central inventory, approval process, centre of excellence.
Security and compliance	Easy building can hide weak access control and data exposure.	Role-based access, reviews, testing and audit trails.
Quality of AI-generated output	Generated logic can contain errors that look correct.	Human review, test cases, version control.
Vendor lock-in	Apps depend on one platform's tools and pricing.	Keep logic in APIs and data exportable; learn transferable skills.
Changing job market	Routine build tasks may shrink as AI improves.	Move toward design, integration, governance and AI oversight.

3.9 Likely future workflow

In the likely future workflow a business need is described in plain language, AI drafts the application, humans review logic, data and security, the application is released through governance, and monitoring feeds new improvements. Human judgement stays at the review and governance steps.

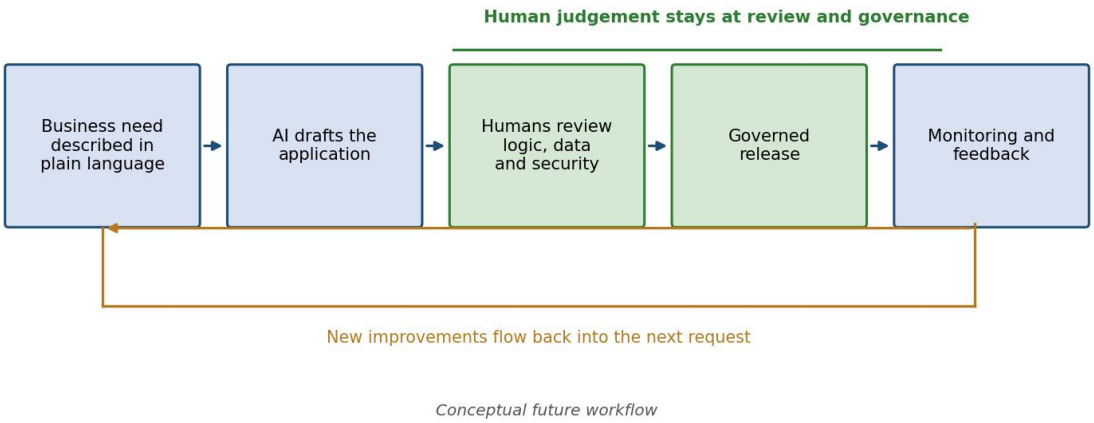


Figure 5: Likely Future Development Workflow (Conceptual Diagram)

3.10 Presentation outline

The findings can be presented in ten slides, one idea per slide:

Slide	Title	Key points
1	Title and introduction	Topic, name and roll number.
2	What is low-code?	Visual building with the option to add code; link to the claim management system.
3	How development evolved	Timeline from hand coding to AI-assisted building.
4	Platform landscape	Zoho Creator, Power Apps, OutSystems, Mendix, Appian, Salesforce.
5	Future trends	AI-assisted building, agents, governed citizen development, hyperautomation.
6	Career pathways	Pathway diagram from entry to senior roles.
7	Skills and roadmap	Skill stack and learning roadmap.
8	Risks and challenges	Shadow IT, security, AI output quality, lock-in.
9	My experience	Experiments 6, 7, 8, 10 and 14 as practical examples.
10	Conclusion and Q&A	Low-code and pro-code will work together; keep learning.

4. Final Result

The study showed that low-code is moving from a faster way to build simple apps to a wider field where visual building, pro-code extension and AI work together. The platform landscape is broad, and the skills learned on one platform such as Zoho Creator transfer to others.

Career pathways run from citizen developer and junior low-code developer, through integration and automation roles, to architect, platform lead and AI-automation roles, with routes into pro-code engineering and product work. The skill stack and roadmap show how the earlier experiments fit into this path.

The main risks are shadow IT, security, the quality of AI-generated output and lock-in, and they are managed by governance and human review. The exercise met its objective, and the presentation outline is ready to be turned into slides. The conclusions are based on general industry knowledge, so any statistics should be checked against the original sources before use.